Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/31/2012
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; STS-135 ; Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 5/7/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roby, Katherine (University of Kansas Medical Center) Gupta, Vijayalaxmi (University of Kansas Medical Center) Holets, Lesya (University of Kansas Medical Center)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Published data demonstrates that short term microgravity (μG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based μG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to μG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term μG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to μG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30d of μG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term μG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to μG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. The observations are also significant in that the results were verified on three separate Space Shuttle experiments (STS-131, STS-133, and STS-135). If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process.
Task Progress:	Summary of histopathological changes in ovary and uterus from STS-131, STS-133, and STS-135: International Space Station, being a National Laboratory enables continued long term habitation in space and study of long term space exposure on the human body. Various estrogen-regulated physiological systems including bone, muscle, wound repair, and immune functions are compromised in space flight. However, studies on the effect of space-flight on reproductive health in females have been limited. The STS-131 BSP program was the first opportunity to examine the effects of space flight on normal cycling adult ovarian and uterine structure/function in rodents. Ovaries from flight mice (C57BL/6) flown for 15 days were significantly smaller ($P<0.001$) compared to ground controls, had significantly fewer ($P<0.003$) corpora lutea (CL), and most of the growing follicles in flight ovaries were atretic, indicative of blocked estrous cycle. Ground and flight mice had no significant difference in number of uterine glands, but there was a trend toward smaller uteri which correlated with observed lower estrogen receptor alpha (ERa) mRNA levels. We repeated this study on STS-133 on Balb/c mice. We observed that most of the larger growing follicles in flight ovaries were atretic and flight ovaries contained no CL compared to ground control ovaries containing between 3 and 6 CL ($P<0.01$). On STS-135 a larger n, similar to STS-131, provided an excellent opportunity to replicate this study in C57Bl6 as on STS-131. The STS-135 spaceflight study demonstrated that ovaries from flight mice were significantly smaller ($P<0.01$) compared to those in ground controls. However, the uteri again did not show significant change between the two groups ($P<0.45$). Detailed analysis of histopathology, protein, and gene transcription analysis of samples from all 3 flights continues. In summary, our studies on STS-131, STS-133 and STS-135 confirm that space-flight significantly and negatively affects ovarian physiology, subsequently comprising the reproductive potential. The potential impact on other estrogen-regulated systems such as bone, muscle, wound repair and immune functions and whether ERa signaling is a common mechanism underlying space flight effects on these systems needs to be determined. STS 131 Gene regulation changes: Microarray analysis was used to compare gene expression profiles between flight and ground control in mouse uteri. Probes were analyzed by microarray experiment in Affymetrix high-density chip; summary of analysis was performed using Ingenuity systems. 5,523 genes were found to be differentially expressed: 2,633 genes significantly up-regulated and 2,890 down-regulated in flight group vs. ground control (p-value<=0.05). Differentially expressed genes involved in several canonical pathways including protein ubiquitination pathway, signaling pathways, immune responses, and metabolic functions. Estrogen receptor signaling pathway was slightly down-regulated (-1.59 fold) after space flight. In flight group we indicate overexpression genes (+4-15 fold) closely associated with tumorogenesis in uterus and other organs (NFE2L2- nuclear factor (erythroid-derived 2)-like 2, CFLAR -CASP8 and FADD-like apoptosis regulator, SPTBN1, TP53 –tumor protein 53, GNA13- G protein alpha 13), and IL-13RA. Among genes down-regulated (-15-4 fold) after gravity exposure are ADIPOQ-adiponectin, FABP4- fatty acid binding protein 4, CFD-adipsin, RETN- hormone activator, resistin, and genes involved in stress response CYP2E1-cytochrome P450, family 2, subfamilyE,polypept.1, and HSPH1 – heat shock protein 105/110kDa. Many of the altered expressed genes played regulatory role in multiple cellular pathways. Real time RT PCR was performed to evaluate the expression levels of ADIPOQ-adiponectin, HSPH1, TP53. Our results indicated significant down-regulation of ADIPOQ and HSPH1 mRNA in flight group compared to ground control (>4 and 30 fold respectively). We did not identify statistically significant changes in TP53 gene expression after space flight. STS 135 Gene regulation changes: Total RNA was extracted from base control, ground control and flight mouse ovaries and uterus tissues. RNA quality and concentrations were determined by Agilent analyses. We report our preliminary

	results from study of a subset of STS 135 flight and ground animals for expression of estrogen receptors alpha and beta (ERa, ERb, progesterone receptors A and B (Prgr AB, Prg B), multifunctional protein lactoferrin (LF), and corticotrophin-releasing hormone receptor 2 (CRH-2) and other genes. Expression of each gene was measured using quantitative real time RT-PCR. ER alpha and beta mRNA levels were significantly lower in both, uterus and ovary from the flight group compared to ground and base controls. Progesterone AB gene expression was down-regulated in flight mice uterus but not ovaries, while progesterone-B was not consistently altered in flight group animals compared to ground and base controls. Lactoferrin mRNA expression is regulated by ERa and ERb levels and statistical reduced in ovaries during space flight. Uterus LF level was not determined up to the present moment. During space flight CRH-R2 mRNA levels (a marker for stress response) were lower in flight mice ovaries but not uterus. Expression levels of another stress marker, HSPH1, dramatically decreased after space flight compare to ground control. TP53 was slightly higher in flight uterus but this effect was not statistical significant. Protein was extracted from base control, ground control and flight mouse uterus. Western blot for ER alpha, CRH-R2, HSPH1 will be performed.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Holets LM, Gupta V, Roby KF, Tash JS. "STS-131: Identification of altered metabolic pathways and gene expression in mice uterus." Joint Meeting: 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, CA, November 3-6, 2011. Program and abstracts. Joint Meeting: 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, CA, November 3-6, 2011. p. 54. , Nov-2011
Abstracts for Journals and Proceedings	Gupta V, Roby K, Holets L, Tash JS. "STS-131, STS-133 and STS-135: Spaceflight Inhibits Ovarian Follicle Development and Formation of Corpus Luteum in Mice." Joint Meeting: 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, CA, November 3-6, 2011. Program and abstracts. Joint Meeting: 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, CA, November 3-6, 2011. p. 53. , Nov-2011
Awards	Tash JS. "President Elect - American Society for Gravitational and Space Research (ASGSR), November 2011." Nov-2011